

Re-evaluating Deep Learning for Lung Cancer Diagnosis

Julia Świątkowska

Abstract

This paper presents a reproduction of the study by Mohammad Q. Shatnawi, Qusai Abuein, and Romesaa Al-Quraan, entitled “Deep Learning-Based Approach to Diagnose Lung Cancer Using CT-Scan Images”, published in *Intelligence-Based Medicine* in 2025. The original study explored the use of several deep learning architectures, including a custom-designed Enhanced Convolutional Neural Network (CNN). For this reproduction, ResNet50 and the Enhanced CNN were selected for focused reimplementations, as their reported performance and architectural choices warranted closer examination. Notably, the Enhanced CNN was claimed to achieve 100% testing accuracy—an assertion that strongly suggests overfitting or methodological issues, particularly given the lack of detailed validation procedures.

In this reproduction, the original methodology is critically re-evaluated. To improve transparency and assess the robustness of the proposed models, explainability techniques such as Grad-CAM were incorporated to visualize model decision-making. The implementation process was carefully reconstructed based on the methods described in the original publication, despite the absence of publicly available source code. Performance metrics were reassessed under varied evaluation conditions to test model reliability. Discrepancies between the reported and reproduced results are discussed, underscoring the challenges of reproducibility in deep learning research. This paper highlights the importance of open science practices—including the sharing of code and data—to facilitate independent validation and foster trust in AI-based medical diagnostics.

1. Introduction

In recent years, artificial intelligence (AI) has increasingly focused on identifying input-output relationships through dynamic resistance signals, moving away from traditional models that rely on predefined assumptions. Modern AI systems are particularly effective at capturing complex, nonlinear interactions. With advances in digital data acquisition, AI now plays a significant role in domains once dominated by human experts, including clinical decision-making, predictive healthcare, patient data analysis, and health system management [1].

AI is especially impactful in the diagnostic and predictive analysis of medical images; however, its integration into clinical practice requires thorough validation. This includes robust performance evaluation, careful consideration of overfitting and bias, and, ultimately, confirmation of clinical utility through well-designed trials and outcome studies [2].

Three key areas demand attention. First, performance validation is critical. For example, a model trained on high-resolution CT scans from one institution may perform poorly when applied to lower-quality images from another, underscoring the importance of external validation and dataset diversity [3]. Second, model interpretability remains a major challenge. The opaque, “black-box” nature of many deep learning

models raises concerns about transparency and accountability. Clinicians must be able to understand the rationale behind AI-generated predictions, particularly in high-stakes scenarios such as cancer diagnosis [4]. Third, there is a significant lack of high-quality clinical trials. To date, only 10 randomized clinical trials on deep learning in medicine have been identified—two of which were published with a low risk of bias (aside from lack of blinding), while eight remain ongoing. Among 81 non-randomized studies, most demonstrated high risk of bias, limited access to data and code, small comparator expert groups, and suboptimal adherence to reporting standards. Nevertheless, the majority of these studies reported that AI performance was comparable to or exceeded that of clinicians, often without adequate validation [5].

This paper examines these pressing challenges through the lens of a recent study that reported 100% accuracy in lung cancer diagnosis using CT scans and tries to replicate the code as accurately as possible since authors didn't provide it [1]. While such results highlight the extraordinary promise of AI in healthcare, they also emphasize the urgent need for rigorous clinical validation, transparent methodologies, and reproducible research practices before widespread clinical adoption can be justified.

2. Methods and Materials

2.1. Data loading

The dataset used in this study, titled “Chest CT-Scan Image Dataset,” was obtained from Kaggle and contains 1,000 CT scan images of lung cancer in .png and .jpg formats. It comprises four classes: Adenocarcinoma, Large Cell Carcinoma, Squamous Cell Carcinoma, and Normal Cells. The images are divided into training, testing, and validation folders.

Due to its relatively small size, the dataset may be insufficient for training deep learning models, which typically require larger datasets to achieve generalizable results. To enhance data organization and prevent potential data leakage, I implemented an extraction method to identify patient IDs from image filenames. This step allows grouping of images by patient, as the dataset contains multiple slices per individual. Organizing the data at the patient level supports more reliable analysis and enables calculation of meaningful statistics, such as the number of unique patients in the dataset.

2.2. Images enhancing

The authors applied several image enhancement techniques (Figure 1.) to improve the quality of the CT scan images prior to further processing. Raw medical images are often of low quality, and enhancement steps help reduce noise, improve contrast, and optimize overall image clarity.

The methods used in this step include:

- Median Filtering (MF): a noise reduction technique that preserves edges while smoothing the image.
- Histogram Equalization (HE): enhances contrast and brightness, improving the visibility of image details.
- Contrast Limited Adaptive Histogram Equalization (CLAHE): divides the image into smaller regions and applies localized histogram equalization, improving local contrast and highlighting finer details.
- Morphological Operations (MO): applied after CLAHE to further enhance image structures and refine contrast, especially useful for highlighting shapes and boundaries in medical images.

These preprocessing steps are essential for improving the input data quality, which in turn supports better performance and reliability of deep learning models.

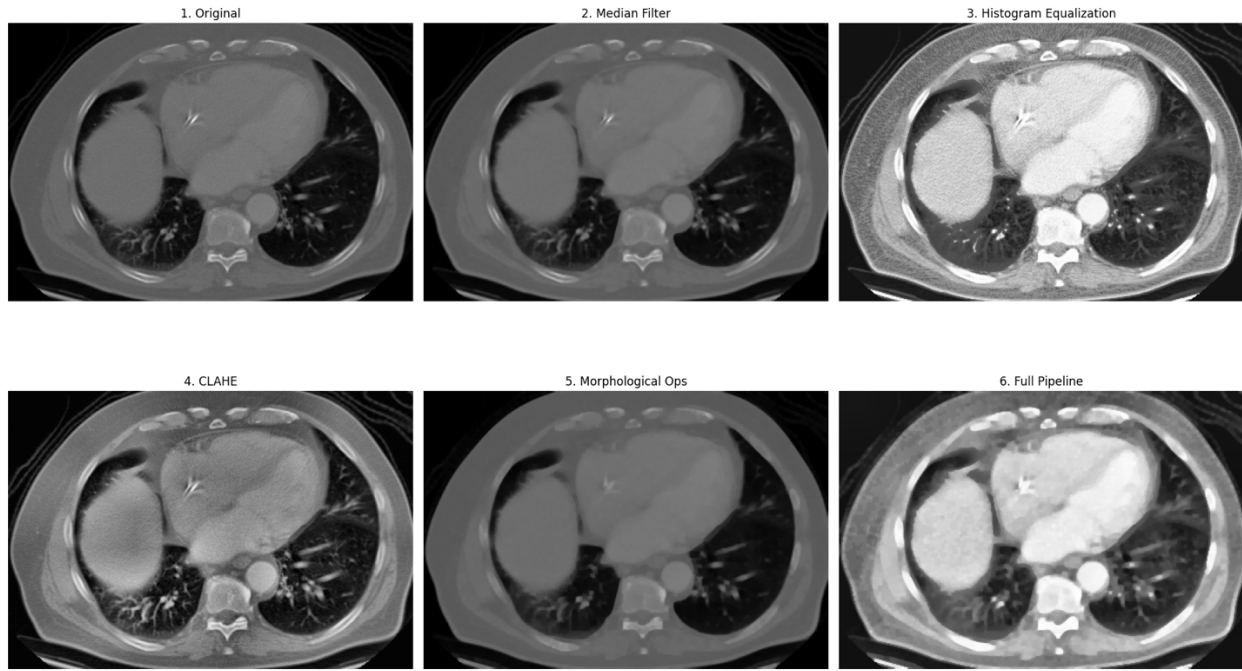


Figure 1. Images enhancement techniques.

2.3. Data pre-processing

Data augmentation was applied to increase the diversity and complexity of the training data, thereby improving the model's ability to generalize to new, unseen cases. A complete augmentation pipeline was implemented and applied to the previously loaded dataset.

The augmentation techniques used in this step include resizing, shear transformation, horizontal flip, vertical flip, and rescaling. These transformations help simulate variability in the data, reduce overfitting, and improve the robustness of the deep learning model during training.

2.4. Models building

2.4.1. Enhanced CNN model

Convolutional Neural Networks (CNNs) are a class of deep learning architectures specifically designed for processing and analyzing visual data. They have demonstrated outstanding performance across a range of computer vision tasks, including image classification, object detection, and particularly in medical imaging applications such as tumor detection and disease classification [6].

The custom CNN model proposed in this study adopts a relatively straightforward architecture. It begins with a series of convolutional layers employing 3×3 filters and ReLU activation functions to extract local spatial features from the input images. Each convolutional layer is followed by a max-pooling layer to progressively reduce the spatial dimensions and computational complexity. To enhance the model's

generalization ability and prevent overfitting, dropout layers are strategically incorporated throughout the architecture.

Following the convolutional and pooling layers, the model transitions into a classification block consisting of fully connected (dense) layers. These layers further abstract the learned features before feeding into the final output layer. The network concludes with a SoftMax-activated dense layer containing four output neurons, corresponding to the four target classes for multi-class lung abnormality classification.

The model was trained using the Adam optimizer and the categorical cross-entropy loss function for 150 epochs, enabling efficient convergence and training with early stopping if no improvement was observed after 20 epochs.

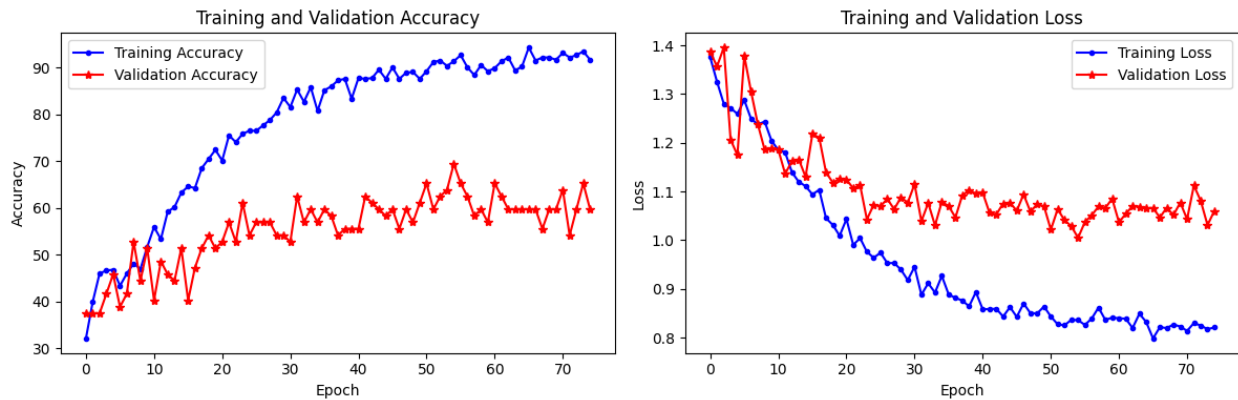


Figure 2. The Enhanced CNN model Accuracy and Loss during training and validation.

2.4.2. ResNet50 model

Residual Neural Network (ResNet50) is a deep convolutional neural network comprising 50 layers. ResNet revolutionized deep learning by addressing the vanishing gradient problem. Instead of learning direct feature mappings, the model learns residuals—the differences between the input and the desired output of a layer. This is achieved through shortcut (or skip) connections that bypass one or more layers, effectively connecting the input of a given layer to a deeper layer. These residual connections enable the model to train deeper architectures more effectively, improving accuracy and convergence in complex tasks like image classification [7].

In the original paper ResNet50 was employed as a pre-trained model to classify lung cancer from CT scan images. It integrates convolutional layers for feature extraction, a pooling layer for spatial reduction and regularization, and fully connected layers for classification. While the authors trained their model using binary cross-entropy loss with the Adam optimizer over 150 epochs and early stopping if no improvement was observed after 20 epochs (splitting data into 70% training and 30% evaluation), several key modifications were made in this implementation to improve robustness and alignment with the multi-class task:

1. Loss function selection:
 - Paper's Approach: Binary cross-entropy (BCE) loss was reportedly used, paired with a SoftMax output layer—a contradictory combination, as BCE requires sigmoid activations for binary/multi-label tasks, while SoftMax is used for mutually exclusive multi-class classification.

- This Implementation: Categorical cross-entropy was adopted instead, as it natively integrates SoftMax and is the correct choice for the 4-class classification task (adenocarcinoma, squamous cell carcinoma, large cell carcinoma, and normal). This ensures proper gradient propagation and probability calibration across mutually exclusive classes.
2. Data splitting:
- Paper's Approach: The original validation set (10%) was combined with the test set (20%), resulting in 30% of the data being used for evaluation. This approach risks inflating test metrics due to potential indirect model tuning on data that was initially intended for validation but later used for testing.
 - This Implementation: Kept the validation set (10%) strictly separate from the test set (20%) to ensure: unbiased early stopping (validation) and clean performance reporting (test) and compliance with reproducibility standards by preventing data leakage.

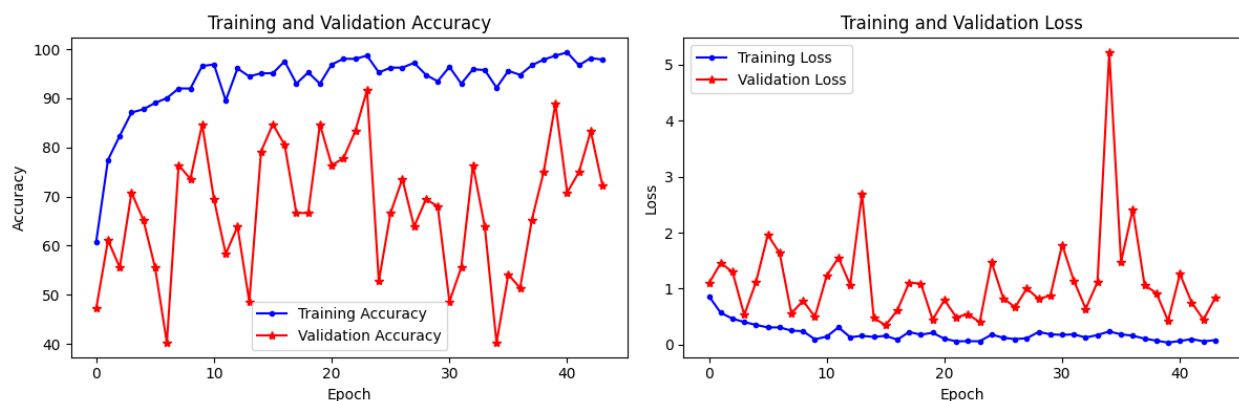


Figure 3. The ResNet50 model Accuracy and Loss during training and validation.

2.5. Grad-CAM – visual explanation technique

As AI plays a larger role in clinical decision-making, interpretability has become crucial for trust in deep learning models, particularly in medical imaging. While CNNs achieve high accuracy, their complexity creates a "black-box" problem, hindering clinical adoption.

To address this, Grad-CAM (Gradient-weighted Class Activation Mapping) leverages the gradients flowing into the last convolutional layer of a CNN—where deeper representations capture high-level semantics while retaining spatial detail—to generate class-discriminative visual explanations. Unlike fully connected layers, which lose spatial information, the final convolutional layers strike a balance between semantic meaning and localization. Grad-CAM assigns importance values to each neuron in this layer, highlighting image regions (e.g., lesions or tumors) most relevant to the model's decision without requiring architectural changes or retraining. This makes it particularly attractive for integration with both pre-trained and custom CNN models used in clinical workflows.

Compared to pixel-level methods like Guided Backpropagation, which can produce visually detailed but class-agnostic maps, Grad-CAM's heatmaps are explicitly class-specific, offering direct insight into how individual features influence a particular classification outcome. These visual explanations not only help validate the model's reasoning but also provide clinicians with intuitive cues that align with established diagnostic practices.

By combining interpretability with strong predictive performance, Grad-CAM bridges the gap between model transparency and clinical utility in medical imaging [8], promoting safer and more explainable deployment of AI systems in sensitive healthcare environments.

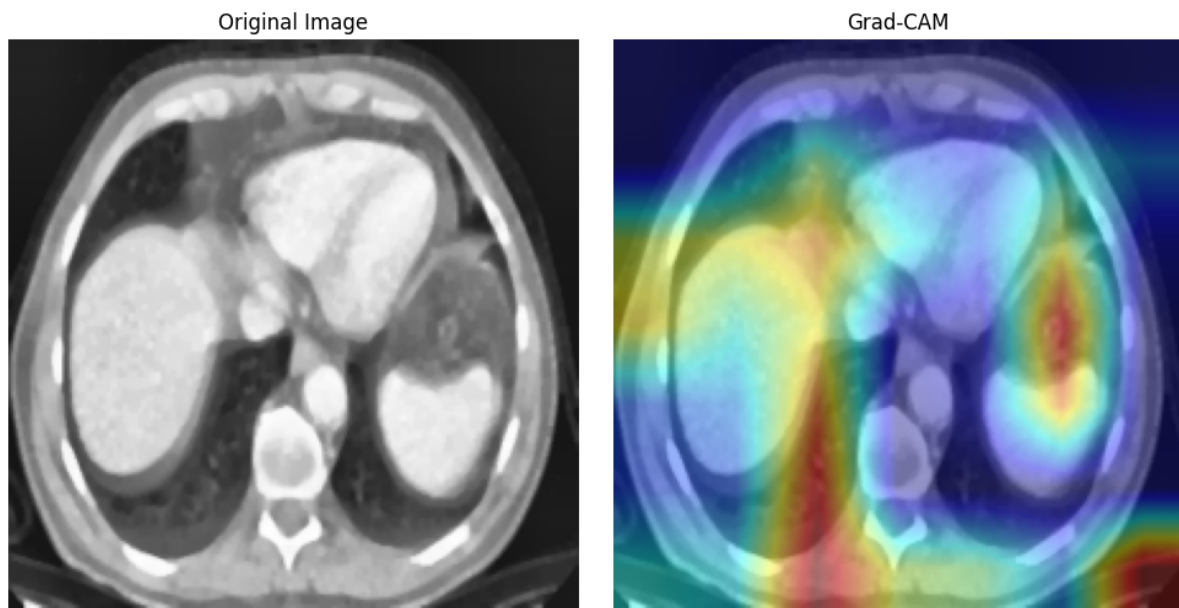


Figure 4. Grad-CAM heatmap for sample image for ResNet50 model.

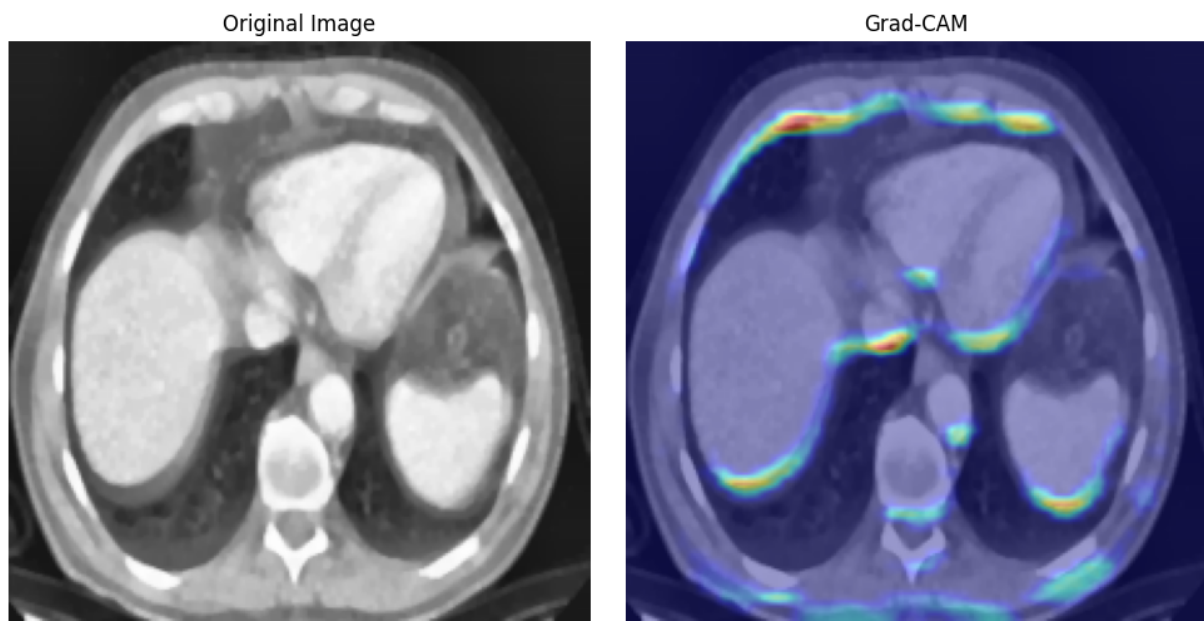


Figure 5. Grad-CAM heatmap for sample image for Enhanced CNN model.

2.6. Performance evaluation metrics

The performance of the reproduced models was evaluated using standard classification metrics, including accuracy, precision, recall, and F1-Score. These metrics provide a comprehensive assessment of

each model's predictive capability, especially in the context of medical imaging where false positives and false negatives can carry significant clinical implications. By analyzing these metrics, it was possible to gauge not only overall correctness (accuracy) but also the models' ability to correctly identify positive cases (recall), avoid false alarms (precision), and maintain a balanced performance across classes (F1 score).

3. Results

The original study reported exceptional performance, including 100% testing accuracy for the Enhanced CNN model. However, closer examination reveals methodological inconsistencies that likely inflated these results. Both the Enhanced CNN and ResNet50 models exhibited clear signs of overfitting, with the validation/test accuracy curve surpassing testing accuracy—an anomalous pattern that suggests evaluation flaws. The authors explicitly stated that they combined the validation and test sets for final evaluation, meaning the model was assessed on data it had already encountered during hyperparameter tuning. This approach artificially boosts performance metrics, as the model was indirectly optimized on portions of the evaluation data.

This reimplementaion adhered to standard practice by strictly separating the validation set (used only for training-phase decisions) and the test set (reserved for final evaluation). While some overfitting persists—evidenced by the expected gap between training and test accuracy—our results reflect a more reliable assessment of generalization. All reported metrics (accuracy, precision, recall, F1-score) were computed exclusively on unseen test data (20% of the dataset), avoiding the circularity present in the original work. The discrepancies between results and the original claims underscore the importance of proper dataset partitioning in machine learning evaluation.

The ResNet50 model achieved an accuracy of 36,19% (94,5% in the original paper), so huge drop in this reimplementaion, which aligns for instance with combination of validation and test sets, which artificially inflates performance metrics. Precision 63,02% indicates the model false positive predictions than claimed. Recall 51,60%, reported sensitivity meaning nearly half of true cancer cases were missed. confirms fundamental limitations in the model's diagnostic capability. These results demonstrate that the architecture, as implemented, fails to meet minimum thresholds for reliable medical diagnosis, highlighting both the challenges of lung cancer detection from CT scans and the potential over-optimism in existing literature about deep learning's readiness for this clinical task. The differences between this results and the original paper's claims suggest readers should interpret the original 94.5% accuracy with caution.

The results from the enhanced CNN model also differed significantly from those reported in the original paper. Where the authors reported perfect 100% accuracy, strictly controlled evaluation from the reimplementaion achieved only 36.83% accuracy on unseen test data. The model's precision of 38.50% and recall of 41.11% (yielding an F1-Score of 36.23%) demonstrate significant challenges in both avoiding false positives and capturing true positives. Several factors likely contribute to this performance gap.

To further evaluate model interpretability, Grad-CAM (Figure 4. and 5.) was applied to visualize the decision-making process of both the ResNet50 and Enhanced CNN models. The resulting heatmaps revealed significant shortcomings in both architectures. Instead of focusing on clinically relevant lung regions or identifiable lesions, the models often attended to irrelevant anatomical structures or diffuse, non-diagnostic areas such as the chest wall or background tissue. This misalignment between highlighted regions and expected pathology suggests that the models failed to learn meaningful medical features and may be relying on spurious correlations.

4. Discussion

The substantial performance discrepancies between the original study and this reimplementaion highlight several critical concerns in the application of deep learning to medical imaging. The most striking issue is the dramatic drop in accuracy—from a reported 100% for the Enhanced CNN and 94.5% for ResNet50 to 36.83% and 36.19%, respectively. These findings strongly suggest that the original results were likely inflated by methodological flaws, particularly the improper combination of validation and test sets or the use of binary cross-entropy loss in a multi-class classification setting.

Such practices, undermine the credibility of model performance claims and pose serious risks when translated into clinical environments. When validation and test sets are not kept separate, performance estimates are no longer indicative of generalization, leading to models that may perform well on paper but fail in real-world scenarios.

5. Acknowledgments

Language and stylistic improvements in this manuscript were aided by OpenAI's ChatGPT.

6. References

- [1] Shatnawi, M. Q., Abuein, Q., & Al-Quraan, R. Deep learning-based approach to diagnose lung cancer using CT-scan images. *Intelligence-Based Medicine* 2025, 11, 100188.
- [2] Park SH, Han K. Methodologic Guide for Evaluating Clinical Performance and Effect of Artificial Intelligence Technology for Medical Diagnosis and Prediction. *Radiology*. 2018 Mar;286(3):800-809.
- [3] Zech JR, Badgeley MA, Liu M, Costa AB, Titano JJ, et al. Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study. *PLOS Medicine* 2018, 15(11), e1002683.
- [4] Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., Batra, D. Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization. *IEEE International Conference on Computer Vision (ICCV)* 2017, 618-626.
- [5] Qi Y, Mohamad E, Azlan AA, Zhang C. Utilization of artificial intelligence in clinical practice: A systematic review of China's experiences. *Digital Health* 2025, 11.
- [6] Mohammed, F.A., Tune, K.K., Assefa, B.G., Jett, M., Muhie, S. Medical Image Classifications Using Convolutional Neural Networks: A Survey of Current Methods and Statistical Modeling of the Literature. *Machine Learning and Knowledge Extraction* 2024, 6(1), 699-735.
- [7] Kamal, K., Ez-Zahraouy, H. A comparison between the VGG16, VGG19 and ResNet50 architecture frameworks for classification of normal and CLAHE processed medical images. *Research Square* 2023.
- [8] Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., Batra, D. Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization. *International Journal of Computer Vision* 2020, 128(2), 336-359.